

Lab 1 — Finding Similar Things

Friday, September 18 (short — same day as Quiz 3)

Last updated: September 13, 2026.

Short lab (~15 minutes after Quiz 3). Full credit is the one-pager check-in. Date is on the [schedule](#).

Credit: Lab points are for effort in class, not for a Canvas file upload. Work in Octave or Python. Handwrite on the [one-pager](#) hard copy and get initials after a short check-in.

Resources: [Lecture 07](#) (norm, distance); [Lecture 08](#) (angle); Lachniet Ch. 1 — [Resources](#).

What we are doing

Practice Euclidean distances in software and name a nearest neighbor — the same skill as Quiz 3 items C1.1–C1.2.

Learning objectives

1. Compute several distances with Octave or Python.
2. Identify a nearest neighbor among a short list of candidates.
3. Notice that rescaling one feature can change the nearest neighbor.

Key syntax

Task	Octave	Python
Norm	<code>norm(x)</code>	<code>np.linalg.norm(x)</code>
Distance	<code>norm(x-y)</code>	<code>np.linalg.norm(x-y)</code>

Exercises

Query $x = (2, 1)$ and candidates

$$z_1 = (2, 2), \quad z_2 = (0, 1), \quad z_3 = (3, 1), \quad z_4 = (2, 0), \quad z_5 = (1.5, 1.5).$$

E1. Compute $\|x - z_i\|$ for $i = 1, \dots, 5$. Which candidate is nearest?

Octave:

```
x = [2; 1];
Z = [2 2; 0 1; 3 1; 2 0; 1.5 1.5]; % row i is z_i
d = sqrt(sum((Z - x').^2, 2))
[~, j] = min(d) % j is 1-based
```

Python:

```
import numpy as np
x = np.array([2.0, 1.0])
Z = np.array([[2, 2], [0, 1], [3, 1], [2, 0], [1.5, 1.5]])
d = np.linalg.norm(Z - x, axis=1)
j = np.argmin(d) # 0-based index
```

Expected: distances 1, 2, 1, 1, $\sqrt{1/2}$; nearest z_5 (Octave j=5, Python j=4).

E2. Rescale the first feature by 10: replace every first coordinate a by $10a$ (so x becomes $(20, 1)$, and likewise for each z_i). Recompute the nearest neighbor(s). Did the answer change?

Expected: distances 1, 20, 10, 1, $\sqrt{25.25}$; nearest is a **tie** between z_1 and z_4 (both distance 1). Yes — the nearest neighbor changed.

One-pager

Preview the [Lab 1 ticket](#); write on the hard copy in class.